



Friedrich-Alexander-Universität
Erlangen-Nürnberg

Gamma Spectroscopy

[ALC - B18]

M766

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Abstract

In this study, we examine gamma photons generated by natural radioactive processes utilizing scintillation and semiconductor detection technologies. The primary aim is to explore and contrast the underlying detection principles of two detector types: a NaI(Tl) scintillation detector and a Ge semiconductor detector. Radioactive sources such as ^{22}Na , ^{60}Co , and ^{137}Cs are employed to produce gamma emissions for analysis. The resulting signals and energy spectra are systematically studied to highlight the influence of both the detectors and the gamma-ray sources on the spectral characteristics. By presenting the data in the form of energy distributions, this experiment offers a detailed understanding of the physical mechanisms that govern the detection of gamma photons in scintillation and semiconductor systems.

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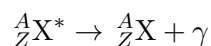
1 Introduction

1.1 γ -Rays

Gamma rays are a form of high-energy electromagnetic radiation typically emitted during nuclear reactions or the radioactive decay of atomic nuclei. Being massless and electrically neutral, gamma rays travel at the speed of light and carry energy that is directly proportional to their frequency (ν), as expressed by the equation:

$$E = h \cdot \nu \quad (1.1)$$

Here, h represents Planck's constant ($6.626 \times 10^{-34} \text{ J} \cdot \text{s}$). Gamma decay is a process in which an unstable nucleus releases excess energy by emitting a gamma photon (γ), transitioning to a more stable configuration. Unlike other forms of radioactive decay, this process does not alter the atomic number or mass number of the nucleus. The general representation of gamma decay is:



where ${}^A_Z\text{X}$ is the original nucleus, ${}^A_Z\text{X}^*$ denotes the excited state of the resulting nucleus, and γ is the emitted gamma ray.

1.2 Interaction of Gamma Rays with Matter

Gamma rays interact with matter through three primary processes:

1. **Photoelectric Effect:** In the low-energy gamma regime, the photoelectric effect is the dominant interaction. In this process, the gamma photon transfers its energy to an inner-shell electron, ejecting it from the atom. The energy of the ejected electron is determined by the difference between the incident gamma photon energy and the binding energy of the electron. This interaction results in the full-energy peak (C) of the gamma spectrum. The photoelectric effect is most dominant at low energies, typically below 100 keV for most materials. For instance, it is significant for Carbon (C) below approximately 10–20 keV,

Aluminium (Al) below 30–50 keV, Iron (Fe) below 50–60 keV, Copper (Cu) below 50–70 keV, Lead (Pb) below 100 keV, and Uranium (U) below 150 keV.

In this interaction, the entire photon energy is absorbed by an atomic electron, which is subsequently ejected from the atom. The cross-section for the photoelectric effect is given by the equation:

$$\sigma_{\text{Photo}} \propto \frac{Z^5}{E_\gamma^{3.5}} \quad (1.2)$$

Here, Z represents the atomic number of the material, and E_γ refers to the energy of incoming photons, typically expressed in keV or MeV. Materials with a high atomic number, such as lead, are particularly effective at absorbing gamma rays due to their stronger interaction with photons. The strong dependence on the atomic number makes high- Z materials highly suitable for shielding against gamma radiation.

2. **Compton Scattering:** Compton scattering is dominant at intermediate gamma energies, typically in the range of 100 keV to 10 MeV. In this interaction, a photon scatters off an electron, losing some of its energy and changing its direction. The cross-section for Compton scattering is given by:

$$\sigma_{\text{Compton}} \propto \frac{Z}{E_\gamma} \cdot \ln(E_\gamma) \quad (1.3)$$

Unlike the photoelectric effect, the cross-section of Compton scattering is less dependent on the atomic number Z . When gamma photons are not fully absorbed, Compton scattering occurs, where the photon transfers a part of its energy to a free or weakly bound electron and is scattered in a different direction. The energy of the scattered photon (E') can be expressed as:

$$E' = \frac{E}{1 + \frac{E}{m_0 c^2} (1 - \cos \theta)} \quad (1.4)$$

In this equation, E' represents the energy of the scattered photon, E is the energy of the incident gamma photon, $m_0 c^2$ is the rest mass energy of the electron, and θ is the scattering angle. When the scattering angle θ is 180° , the photon transfers its maximum possible energy to the electron, producing the Compton edge (B). Scattering at angles less than 180° results in a range of lower photon energies, which forms the Compton continuum (A).

3. **Pair Production:** Pair production occurs at gamma photon energies above 1.022 MeV, which is equivalent to twice the rest mass energy of an electron. In this interaction, the photon is converted into an electron-positron pair in the presence of a nucleus. The cross-section for pair production is given by:

$$\sigma_{\text{Pair}} \propto Z^2 \cdot \ln(E_\gamma) \quad (1.5)$$

This interaction becomes significant at high gamma photon energies and depends strongly on the square of the atomic number (Z^2) of the material, as well as the logarithm of the photon energy (E_γ).

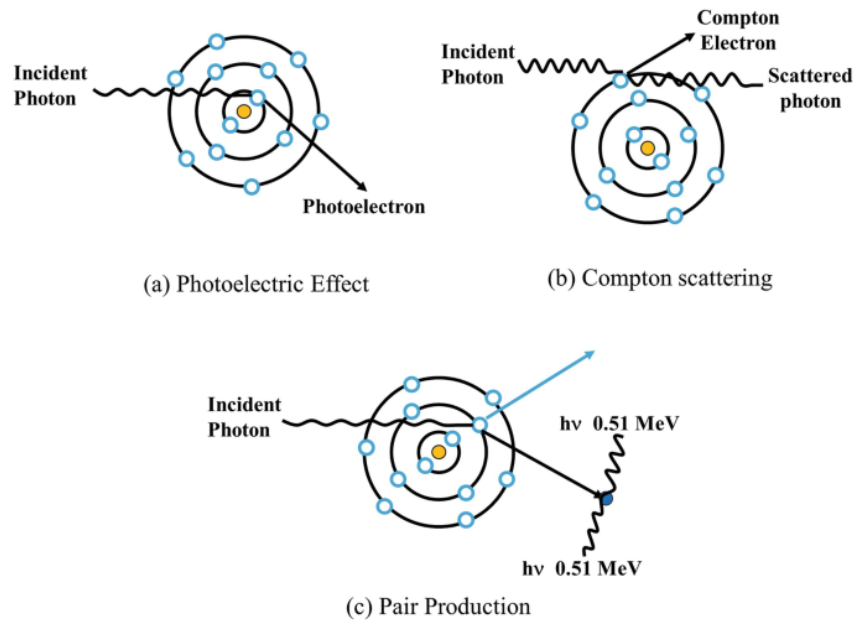


Figure 1.1: Representation of γ interactions with matter

1.3 Absorption Law

The attenuation of gamma rays as they pass through a material is described by the exponential absorption law:

$$I(x) = I_0 e^{-\mu x} \quad (1.6)$$

where:

- $I(x)$ is the intensity of the gamma radiation at depth x ,
- I_0 is the initial intensity of the gamma radiation,

- μ is the linear attenuation coefficient, which depends on the material and the energy of the gamma rays,
- x is the thickness of the absorbing material.

The attenuation coefficient μ is the sum of contributions from the photoelectric effect, Compton scattering, and pair production.

Determining Photon Interaction Cross Section (σ) from μ

The photon interaction cross section (σ) is associated with the linear attenuation coefficient (μ) through the relationship:

$$\mu = n \cdot \sigma \quad (1.7)$$

where n , the number density of atoms in the material, is calculated as:

$$n = \frac{N_A \cdot \rho}{M} \quad (1.8)$$

Here:

- n : Number density of atoms in the material,
- N_A : Avogadro's constant ($6.022 \times 10^{23} \text{ mol}^{-1}$),
- ρ : Density of the material, expressed in kg/m^3 or g/m^3 ,
- M : Molar mass of the material in g/mol .

By rearranging the equation, the cross section (σ) can be expressed as:

$$\sigma = \frac{\mu}{n} \quad (1.9)$$

1.4 Scintillation Detector

Scintillation detectors are essential instruments for detecting and analyzing gamma-ray spectra due to their high sensitivity and efficiency. These detectors operate on the principle of converting the energy of incident gamma photons into detectable light signals, which are then amplified and processed to extract spectral information. The detection process is summarized below, incorporating key interactions and mechanisms.

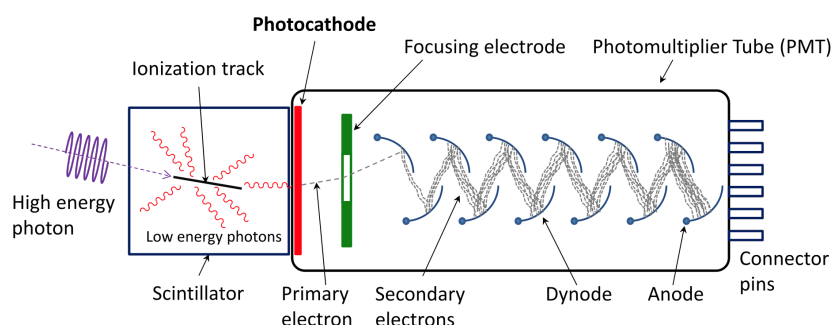


Figure 1.2: Schematic representation of the working principle of a scintillation detector.

1.4.1 Components of a Scintillation Detector

Scintillator Material

A scintillator is a material that converts high-energy radiation, such as gamma rays, into visible or near-visible light through a process called *scintillation*. This process involves absorbing the energy from gamma-ray photons and re-emitting part of it as light.

Materials Used Common scintillator materials include Sodium Iodide doped with Thallium (NaI(Tl)) and Cesium Iodide doped with Thallium (CsI(Tl)). NaI(Tl) is one of the most widely used scintillators due to its high efficiency in converting gamma rays to light. It emits light in the visible range, with a wavelength of approximately 415 nm. In contrast, CsI(Tl) is often preferred in high-radiation environments due to its mechanical robustness, although it has a slightly slower response compared to NaI(Tl). The light emitted by CsI(Tl) is in the visible range, with a wavelength of approximately 550 nm.

Interaction with Gamma Rays When gamma rays interact with the scintillator material, the process involves three main steps. First, the gamma rays transfer their energy to the electrons in the material through interactions such as the photoelectric effect, Compton scattering, or pair production. This step results in gamma-ray absorption. Next, the energy of the gamma rays excites the electrons in the scintillator material to higher energy states. Finally, these excited electrons quickly return to their ground state by releasing energy in the form of light photons, a process known as scintillation. The amount of light emitted is proportional to the energy deposited by the gamma ray.

Photomultiplier Tube (PMT)

The scintillation light generated in the scintillator is transmitted to a photomultiplier tube (PMT), where it is converted into an electrical signal. Inside the PMT, photons strike a photocathode, releasing photoelectrons. These electrons are then multiplied as they travel through a series of dynodes, each maintained at a higher potential, amplifying the signal. If each dynode multiplies the electrons by a factor ξ , and there are D dynodes, the total number of electrons reaching the anode for a single photoelectron is ξ^D . For N primary photoelectrons, the total output becomes $\xi^D \cdot N$. Since N is proportional to the energy deposited by the gamma photon, the PMT output is directly proportional to $\xi^D \cdot E_\gamma$.

Energy Proportionality

The linear relationship between the number of electrons at the PMT anode and the gamma-photon energy allows scintillation detectors to produce energy spectra that accurately represent the gamma-ray interactions.

By analyzing the resulting energy spectra, scintillation detectors provide insights into gamma-ray sources and their interactions with matter. The features of the spectrum, such as the full-energy peak, Compton edge, and continuum, reflect the underlying physical processes occurring during detection. Here, the Compton Edge occurs due to the maximum energy transfer to an electron during Compton scattering, while the Compton Continuum arises from varying energy transfers at different scattering angles.

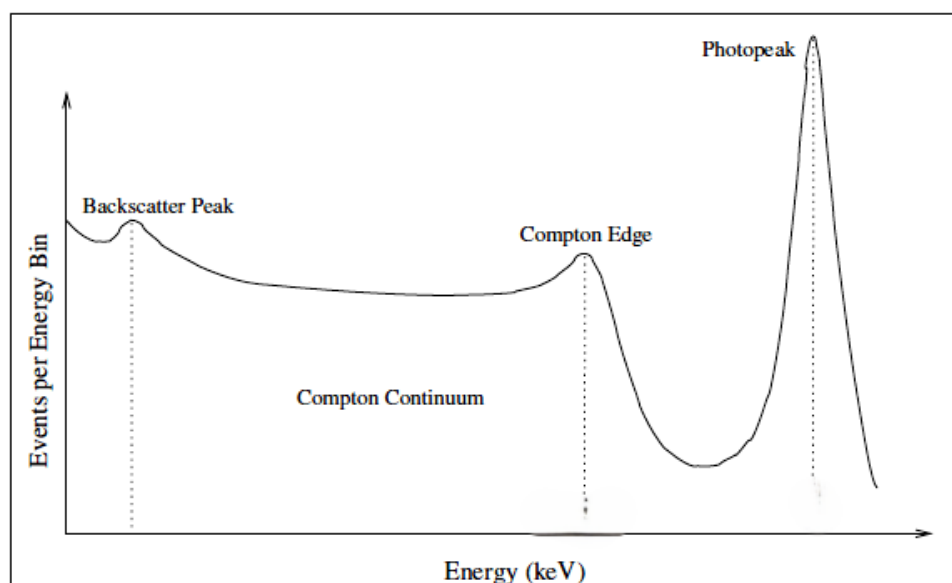


Figure 1.3: Typical gamma-ray spectrum obtained using a scintillation detector, showing key features such as Compton continuum, Compton edge, and photopeak.

1.5 Semiconductor Detector

Semiconductor detectors are a class of radiation detectors that utilize materials like silicon or germanium to detect and measure ionizing radiation. These detectors rely on the electronic properties of crystalline structures, where the arrangement of atoms forms periodic lattices, creating defined energy bands for electrons. Electrons within the semiconductor are restricted to specific energy ranges:

- **Valence band:** Contains electrons bound to atoms in the crystal lattice.
- **Conduction band:** Contains free electrons that contribute to electrical conductivity.

The energy gap between these two bands, known as the band gap (Figure ??), determines the material's electrical properties. Unlike insulators with large band gaps or conductors with negligible gaps, semiconductors have intermediate band gaps, typically around 1 eV. Thermal excitation in a semiconductor can result in the creation of electron-hole pairs, where an electron transitions to the conduction band, leaving a hole in the valence band.

1.5.1 Detection Mechanism

In germanium-based detectors, gamma photons interact with the atoms in the material, often through the photoelectric effect. This interaction ejects an inner-shell electron from the atom, completely absorbing the photon's energy and resulting in a full-energy peak (C) in the gamma spectrum. The energy deposited by the photon generates electron-hole pairs within the detector, which are subsequently collected to form an electrical signal.

Gamma photons can also undergo Compton scattering, transferring only a portion of their energy to electrons within the material. This process creates the Compton edge (B) in the spectrum, which represents the maximum energy transfer from a photon to an electron during scattering. Lower energy transfers, caused by varying scattering angles, produce the Compton continuum (A), which spans a range of energies below the Compton edge.

1.5.2 Semiconductor Types and PN Junctions

Semiconductors play a vital role in radiation detection and are modified by doping them with impurities to create P-type or N-type materials. Combining these two types forms a PN junction diode, which creates a depletion region acting as a particle detection zone. When reverse biased, the depletion region expands, reducing current flow in the absence of radiation. When a gamma photon or other ionizing particle interacts with the depletion region, it generates electron-hole pairs proportional to the particle's energy.

The configuration of electronic states in metals, semimetals, semiconductors, and insulators is shown in Figure 1.4, highlighting the energy band gap (E_g).

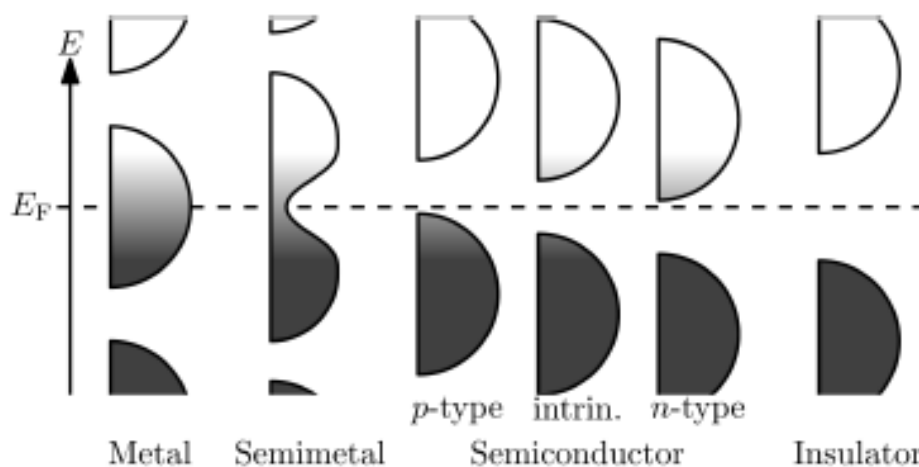


Figure 1.4: Configuration of electronic states in metals, semimetals, semiconductors, and insulators, showing the energy band gap (E_g).

To suppress thermal excitation and maintain detector sensitivity, semiconductor detectors, particularly those based on germanium, are cooled to cryogenic temperatures. The band gap for germanium is approximately 0.7 eV, while for silicon, it is around 1.12 eV. The energy required to produce an electron-hole pair is about 3 eV for germanium and 3.6 eV for silicon. For a 1 MeV gamma photon, approximately 3×10^5 electron-hole pairs are generated in germanium, compared to 2.7×10^5 pairs in silicon.

1.5.3 Energy Resolution

Semiconductor detectors, such as High Purity Germanium (HPGe) detectors, provide superior energy resolution compared to scintillation detectors. This is due to the higher number of electron-hole pairs produced compared to the number of photons produced by a scintillator for particles with similar energies, allowing for more precise energy measurements. The energy resolution of an HPGe detector is about 0.18%, which can be determined approximately using the standard deviation of the Poisson distribution for the number of electron-hole pairs created. However, this provides only a lower estimate of the uncertainty, as it does not account for other factors such as electronic noise, charge collection inefficiencies, or fluctuations in the amplification process. Another measure of energy resolution is the Full Width at Half Maximum (FWHM) of the energy peak, expressed as FWHM/E , where E is the peak energy.

In summary, semiconductor detectors like HPGe offer high precision in gamma-ray spectroscopy, enabling detailed analysis of spectral features and energy distributions. Their advanced resolution and ability to distinguish closely spaced peaks make them invaluable for applications

requiring accurate radiation measurements. While semiconductor detectors are more sensitive and precise, scintillation detectors are preferred in situations where cost, size, ruggedness, or simplicity of operation is more critical. Each detector type has its own strengths, and the choice depends on the specific requirements of the application.

1.6 Activity of Radioactive Substances

Radioactive substances emit radiation as their nuclei undergo decay, a process that occurs at a specific rate. This rate, known as the **activity** of the substance, quantifies the number of decays taking place per unit time. Activity is measured in **becquerels (Bq)**, where one becquerel corresponds to a single decay per second.

The activity A of a radioactive material is governed by the relationship:

$$A = \lambda \cdot N \tag{1.10}$$

Here:

- λ : Decay constant, characteristic of the radioactive material.
- N : Total number of radioactive nuclei currently present.

This formula highlights the direct proportionality between the activity of a substance and the number of undecayed nuclei, emphasizing the dependence of decay rate on both the material's inherent properties and the quantity of nuclei available to decay.

2 Methodology

2.1 Measurement Setup

The experimental setup is schematically illustrated in Figure 2.1. When gamma photons interact with the detector material, energy is transferred to the detector in a primary interaction process. The corresponding output signal from the detector, provided the quantum is fully absorbed, is proportional to the total energy of the photon. These signals are subsequently amplified and processed by an Analog-to-Digital Converter (ADC) integrated into a computer for graphical representation.

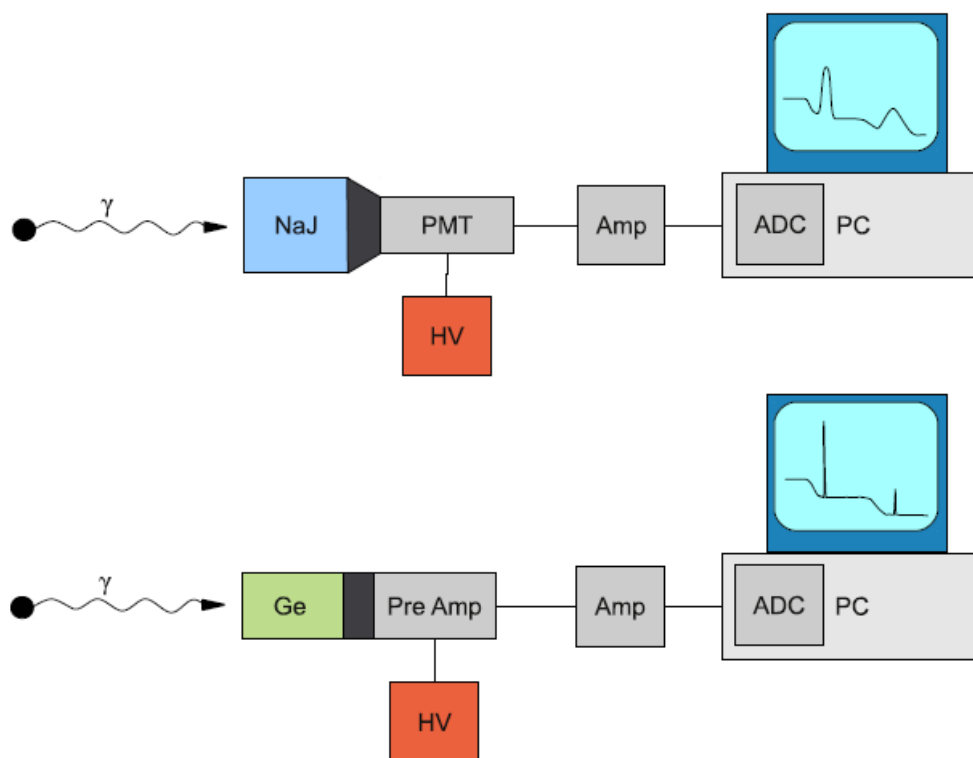


Figure 2.1: Schematic of the experimental setup: top - scintillator detector, bottom - semiconductor detector.

Key components of the setup include:

- **NaI(Tl) Crystal:** A scintillating crystal used for gamma detection.
- **PMT (Photo Multiplier Tube):** Converts scintillation light into electrical signals and amplifies them.
- **HV (High Voltage):** Provides the necessary bias voltage for the PMT or preamplifier.
- **Ge Detector:** A germanium semiconductor detector for high-resolution gamma spectroscopy.
- **Preamp and Amp:** Amplifies the electrical signals for further processing.
- **ADC (Analog-to-Digital Converter):** Converts the analog signals into digital form for analysis on a personal computer.

The table below provides a detailed list of the radioactive sources used in this experiment, including their activity, calibration date, and storage details.

Isotope	Activity (kBq)	Calibration Date	ID	Storage
^{22}Na	440	01.12.1989	BV 773	Wooden Box
^{137}Cs	430	01.12.1989	BV 770	Wooden Box
^{60}Co	426	01.12.1989	BV 771	Wooden Box
^{60}Co	37	01.12.2007	BV 782	Large Lead Cylinder
^{60}Co	18	01.12.2007	PU 149	Case
^{22}Na	374	01.08.2018	AM 1736	Small Lead Cylinder

Table 2.1: List of radioactive sources used in the gamma spectroscopy experiment.

2.2 Decay Schemes of Radioactive Isotopes

2.2.1 Decay Scheme of ^{22}Na

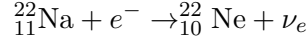
The radioactive isotope ^{22}Na undergoes decay via two primary modes: beta decay (β^+) and electron capture. The decay pathways and branching ratios are depicted in Figure 2.2.

In the beta decay mode, which accounts for 90% of the decays, ^{22}Na emits a positron (e^+) and a neutrino (ν_e) while transforming into ^{22}Ne . The reaction can be expressed as:



Following this process, the daughter nucleus ^{22}Ne is left in an excited state and de-excites by emitting a gamma photon with an energy of 1.274 MeV. This gamma emission is characteristic of ^{22}Na decay.

In the electron capture mode, comprising 10% of the decays, ^{22}Na captures an orbital electron, leading to a transition to ^{22}Ne . This process is represented as:



Similar to the beta decay pathway, the excited ^{22}Ne nucleus de-excites by emitting a gamma photon with an energy of 1.274 MeV.

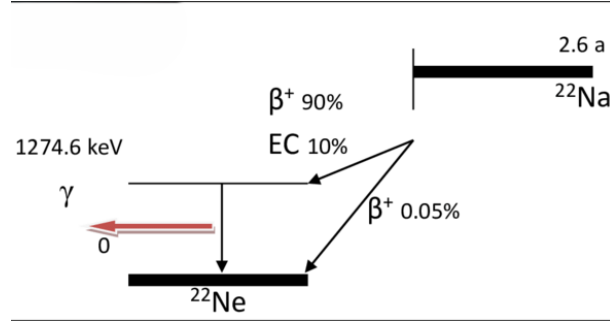
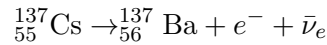


Figure 2.2: Decay scheme of ^{22}Na with branching ratios.

2.2.2 Decay Scheme of ^{137}Cs

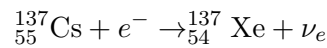
The radioactive isotope ^{137}Cs undergoes decay primarily through beta decay, along with minor contributions from other decay modes.

In the beta decay mode, ^{137}Cs transforms by converting a neutron into a proton, emitting an electron (e^-) and an antineutrino ($\bar{\nu}_e$). The resulting product is ^{137}Ba :



The ^{137}Ba nucleus is produced predominantly in an excited state, which de-excites by emitting a gamma photon with an energy of 0.662 MeV. This gamma radiation is a characteristic signature of ^{137}Cs decay and is widely used in radiation detection and calibration.

Although beta decay is the dominant mode, ^{137}Cs can also undergo electron capture, leading to the formation of ^{137}Xe :



However, this decay pathway has a negligible branching ratio compared to beta decay and does not significantly contribute to the overall radiation spectrum.

The primary decay product of ^{137}Cs is ^{137}Ba . The excited state of ^{137}Ba emits 0.662 MeV gamma rays during its transition to the ground state, making ^{137}Cs a significant source of gamma radiation.

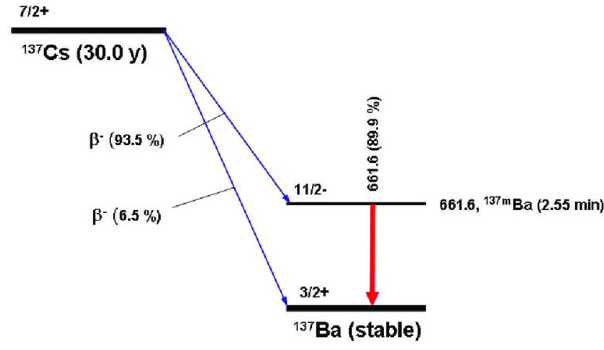
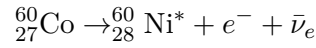


Figure 2.3: Decay scheme of ^{137}Cs with branching ratios and decay pathways.

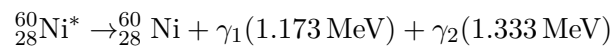
2.2.3 Decay Scheme of ^{60}Co

Cobalt-60 (^{60}Co) is a radioactive isotope that undergoes beta decay to transform into stable nickel-60 (^{60}Ni). During this decay process, the nucleus emits an electron and an antineutrino, transitioning to an excited state of nickel, which then rapidly de-excites by emitting two characteristic gamma rays. The decay is characterized by a half-life of 5.2714 years and involves two prominent gamma rays with energies of 1.173240 MeV and 1.332508 MeV, which are emitted with nearly 100% intensity.

The nuclear equation for the beta decay of ^{60}Co is:



Here, the excited nickel-60 ($^{60}\text{Ni}^*$) nucleus is produced. This nucleus then de-excites by emitting two gamma photons sequentially:



The decay mechanism begins with the ^{60}Co nucleus (5+ spin state) transforming into an excited state of ^{60}Ni at 2.505748 MeV. This excited state quickly transitions through an intermediate energy level at 1.332508 MeV before reaching the ground state of ^{60}Ni (0+ spin state). The beta decay has a maximum energy of 317.32 keV, with a primary transition probability of 99.88%. While the main decay path is well-defined, there are also minor transitions with very low probabilities, including rare beta transitions to other excited states of ^{60}Ni and potential internal conversion processes.

The significance of ^{60}Co extends beyond its fundamental nuclear properties, finding critical applications in medical radiotherapy, industrial radiography, and scientific research as a reliable gamma-ray source for calibration and experimental purposes.

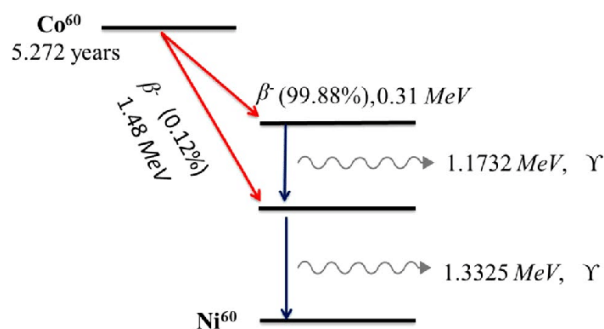


Figure 2.4: Decay scheme of ^{60}Co radionuclide, illustrating the beta decay process and the subsequent gamma transitions.

2.2.4 General Observations on Decay Schemes

Decay schemes such as those of ^{22}Na , ^{137}Cs and ^{60}Co reveal how radioactive isotopes transition to more stable configurations through specific pathways. Each pathway is associated with branching ratios, which quantify the likelihood of each process. Importantly, the sum of all branching ratios equals unity, reflecting the total probability of decay. The energy states and subsequent transitions of decay products contribute to the complexity of radiation spectra observed in gamma spectroscopy experiments.

3 Analysis

3.1 Oscilloscope Signals of the Detectors

3.1.1 NaJ Scintillation Detector

The NaJ scintillation detector was connected to the oscilloscope, and both direct (PMT output) and amplified signals were analyzed. The direct signal exhibited an amplitude of -5.91 V , with a rise time of $1.41\text{ }\mu\text{s}$ and a fall time of $1.22\text{ }\mu\text{s}$. After amplification, the signal showed an amplitude of 2.20 V , with a rise time and fall time of $2.35\text{ }\mu\text{s}$ each.

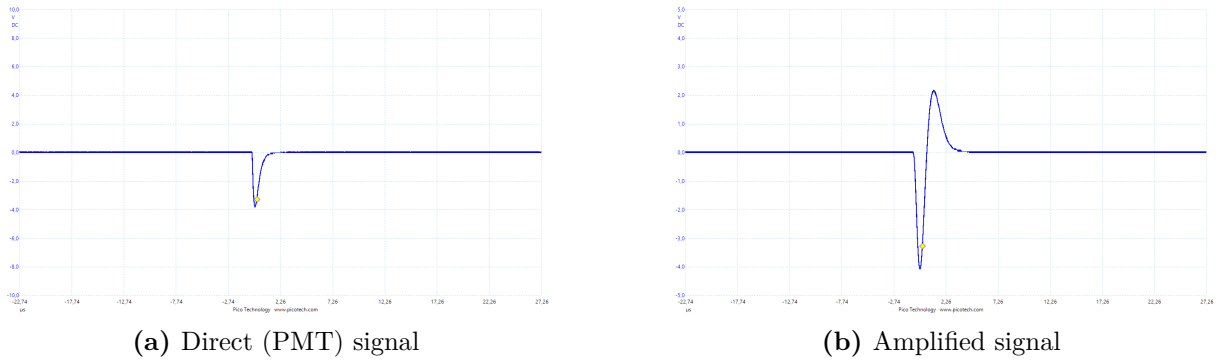


Figure 3.1: Oscilloscope signal for ^{60}Co by the NaJ scintillation detector

The shorter rise and fall times observed in the NaJ scintillation detector indicate its ability to resolve closely spaced events effectively. The amplification enhanced the signal strength, making it more suitable for further analysis while maintaining the original timing characteristics.

3.1.2 Ge Semiconductor Detector

For the Ge semiconductor detector, the direct signal (PMT output) showed an amplitude of 91.47 mV , with a rise time of 5.47 ms and fall time of 4.97 ms . After amplification, the amplitude increased significantly to 2.52 V , and the rise and fall times were measured to be $10.47\text{ }\mu\text{s}$.

The Ge detector exhibited longer rise and fall times compared to the NaJ scintillation detector. These longer times reduce its time resolution and increase susceptibility to pile-up, particularly

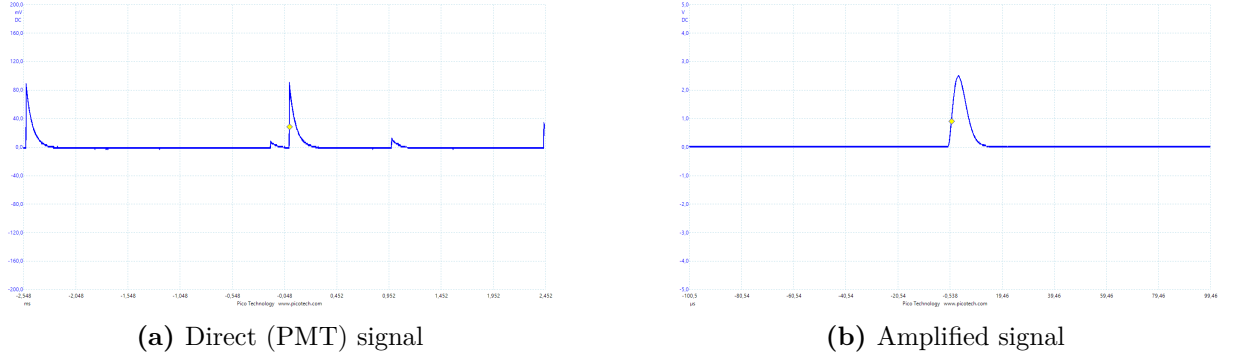


Figure 3.2: Oscilloscope signal for ^{60}Co by the Ge semiconductor detector

at higher event rates. Amplification successfully improved the signal strength for subsequent analysis while slightly broadening the timing characteristics.

3.1.3 Pile-Up

Pile-up occurs when multiple gamma-ray events are detected within a time interval shorter than the detector's resolving time, leading to overlapping signals. This phenomenon was observed more frequently in the Ge semiconductor detector due to its longer rise time, which increases the likelihood of overlapping signals, as shown in Figure 3.3.

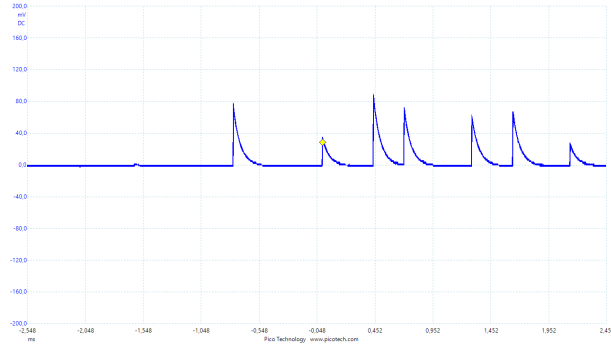


Figure 3.3: Pile-up signal

3.2 Shaping Time and Gain

The output from both detectors was amplified and shaped using tunable shaping times and gain values. The shaping time refers to the duration over which the signal is processed, influencing the pulse width. Shorter shaping times lead to narrower pulses and reduced pile-up but may degrade the signal-to-noise ratio. Conversely, longer shaping times produce broader pulses, increasing the risk of pile-up while enhancing noise suppression.

For the NaJ scintillation detector, shaping time and gain were optimized to $0.5\,\mu\text{s}$, 5 coarse gain, and 6 fine gain, ensuring the expected spectrum and minimal signal distortion. The Ge semiconductor detector required a shaping time of $2\,\mu\text{s}$, with a 50 coarse gain and 12 fine gain, providing a balance between peak visibility and noise suppression.

The gain settings amplified the signal amplitudes for both detectors, improving sensitivity to low-energy events and ensuring effective signal analysis.

3.2.1 Comparison of Time Resolution

The NaJ scintillation detector, with its shorter rise and fall times, demonstrated superior time resolution compared to the Ge semiconductor detector. The ability to distinguish events separated by shorter intervals makes the NaJ detector less prone to pile-up. In contrast, the Ge detector's longer rise time increases the likelihood of overlapping signals, limiting its application in high-event-rate scenarios.

3.3 Energy Calibration and Resolution Analysis

3.3.1 Theoretical Background

Energy calibration in gamma-ray spectroscopy establishes a quantitative relationship between the channel number of the detector's output and the corresponding photon energy in keV. This calibration is essential for identifying unknown gamma-ray energies in spectra obtained from radioactive sources. The process relies on the detector's linear response to the energy of incoming gamma rays, enabling the use of a linear model to map channel numbers to energy values.

For this analysis, the calibration curve is described by the equation:

$$E(\mu) = q \cdot \mu + r \quad (3.1)$$

where:

- $E(\mu)$ is the photon energy (keV),
- μ is the observed channel number,
- q is the slope of the calibration line (keV/channel), representing the detector's energy resolution per channel,
- r is the offset, accounting for systematic deviations in the detector's response.

Energy resolution, another critical parameter, quantifies the detector's ability to distinguish between closely spaced energy peaks. It is defined as:

$$R = \frac{\text{FWHM}}{E} \times 100\% \quad (3.2)$$

where:

- R is the resolution percentage,
- FWHM is the Full Width at Half Maximum of the peak (keV),
- E is the centroid energy of the peak (keV).

The FWHM is derived from the standard deviation (σ) of the Gaussian fit to the peak, given by:

$$\text{FWHM} = 2.355 \cdot \sigma$$

The calibration and resolution analysis presented in this section evaluates the performance of two detectors—NaI(Tl) scintillation and Ge semiconductor detectors—using the spectra obtained from ^{22}Na , ^{60}Co , and ^{137}Cs . These detectors differ in their mechanisms and resolutions, as reflected in the measured calibration curves and energy resolutions.

3.3.2 Spectra Obtained from NaI(Tl) Detector

The NaI(Tl) scintillation detector was used to measure the gamma-ray spectra of ^{22}Na , ^{60}Co , and ^{137}Cs . The detector is widely used due to its high detection efficiency and moderate energy resolution. Gaussian functions were fitted to the peaks corresponding to full-energy peaks (FEPs). The fit parameters are summarized in Table 3.1. These parameters were used to calculate the Full Width at Half Maximum (FWHM) and energy resolution.

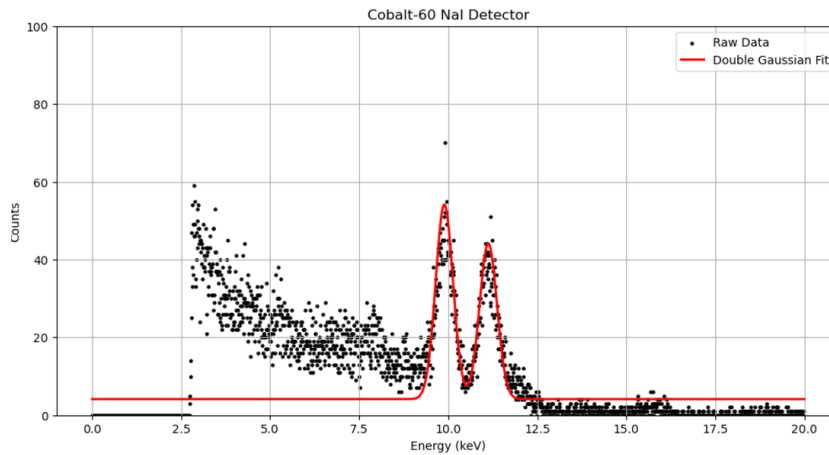


Figure 3.4: ^{60}Co spectrum for NaI detector

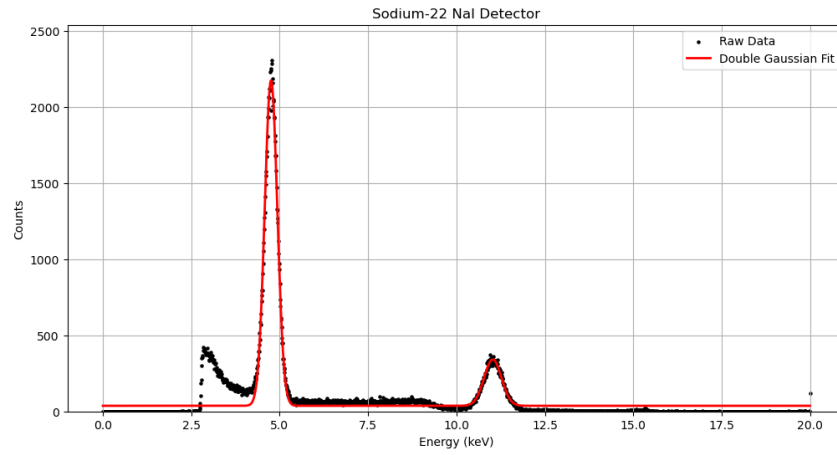


Figure 3.5: ^{22}Na spectrum for NaI detector

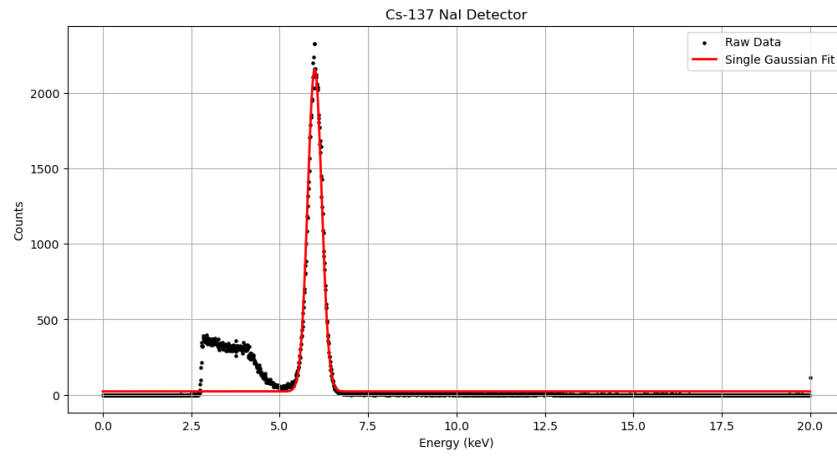


Figure 3.6: ^{137}Cs spectrum for NaI detector

Table 3.1: Gaussian Fit Parameters for NaI(Tl) Detector

Source	Amplitude (a)	Mean (μ)	Std. Dev. (σ)	Baseline
^{22}Na (511 keV)	2139.06 ± 14.34	4.76 ± 0.001	0.18 ± 0.001	37.02 ± 1.56
^{22}Na (1275 keV)	304.46 ± 12.23	11.03 ± 0.011	0.25 ± 0.011	-
^{60}Co (1173 keV)	50.00 ± 2.42	9.89 ± 0.01	0.24 ± 0.01	4.12 ± 0.26
^{60}Co (1332 keV)	40.00 ± 2.20	11.12 ± 0.01	0.25 ± 0.01	-
^{137}Cs (662 keV)	2132.06 ± 85.22	5.99 ± 0.01	0.20 ± 0.01	22.64 ± 2.56

3.3.3 Spectra Obtained from Ge Detector

The Ge semiconductor detector was used to measure the same gamma-ray spectra. Due to its excellent energy resolution, the Ge detector provides narrower and well-separated peaks compared to the NaI(Tl) detector. Table 3.2 summarizes the Gaussian fit parameters for the Ge detector. The narrower peaks contribute to its suitability for applications requiring high precision.

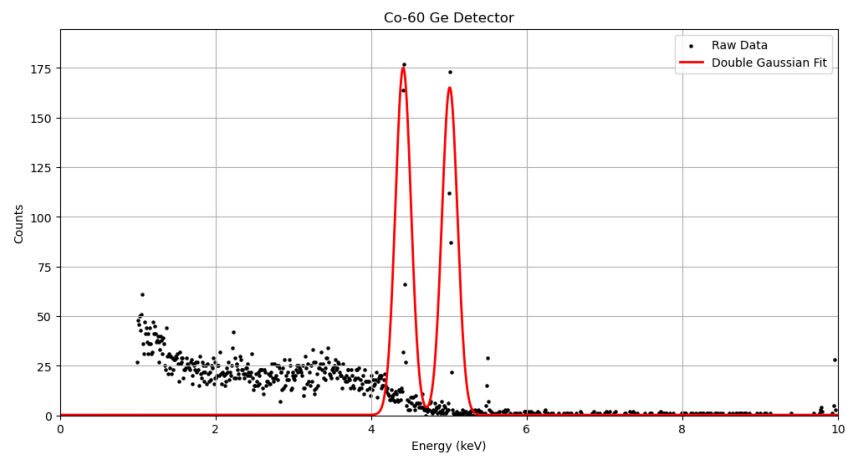


Figure 3.7: ^{60}Co spectrum for Ge detector

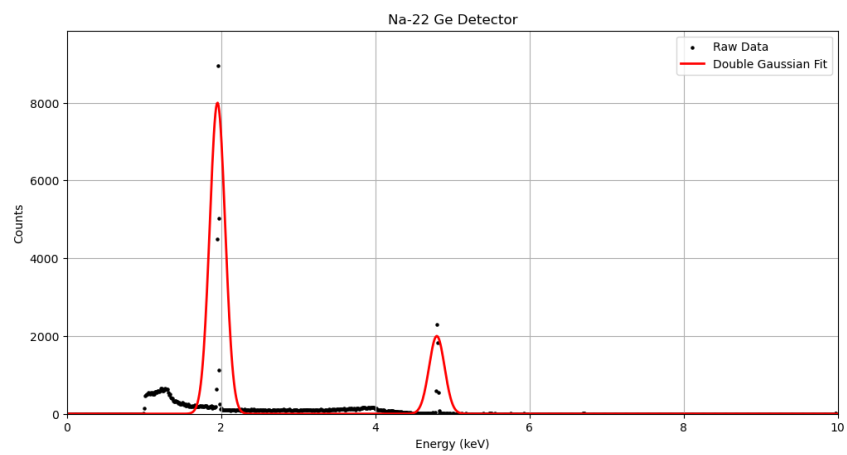


Figure 3.8: ^{22}Na spectrum for Ge detector

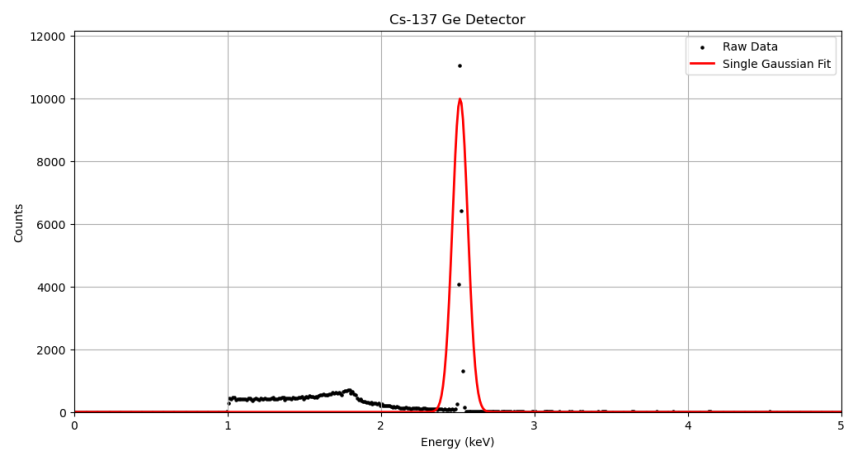


Figure 3.9: ^{137}Cs spectrum for Ge detector

Table 3.2: Gaussian Fit Parameters for Ge Detector

Source	Amplitude (a)	Mean (μ)	Std. Dev. (σ)	Baseline
^{22}Na (511 keV)	8000.00 ± 202.81	1.95 ± 0.003	0.10 ± 0.003	0.00 ± 15.99
^{22}Na (1275 keV)	2000.00 ± 202.81	4.80 ± 0.012	0.10 ± 0.012	-
^{60}Co (1173 keV)	175.00 ± 6.44	4.41 ± 0.004	0.10 ± 0.004	0.29 ± 0.51
^{60}Co (1332 keV)	165.00 ± 6.44	5.01 ± 0.005	0.10 ± 0.005	-
^{137}Cs (662 keV)	10000.00 ± 230.28	2.52 ± 0.001	0.05 ± 0.001	0.00 ± 12.59

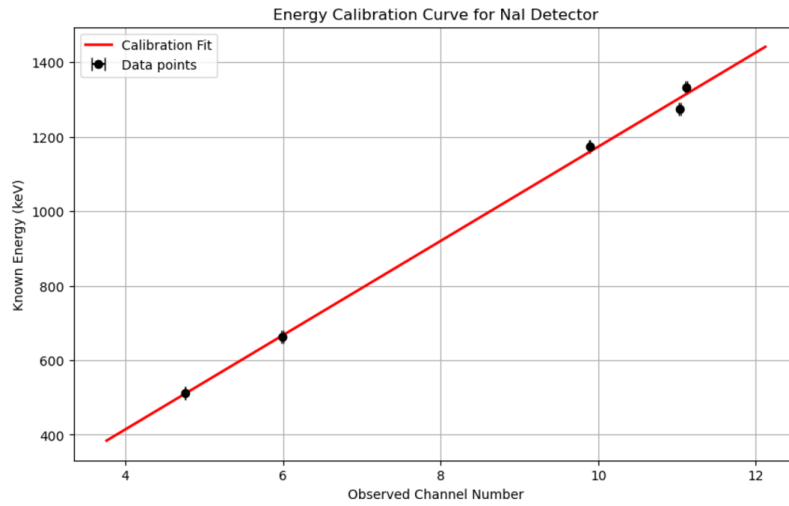
3.3.4 Energy-Channel Calibration

The energy-channel calibration was performed by plotting the known energies of FEPs against their corresponding channel numbers. Linear calibration equations were derived for both detectors:

$$\text{NaI(Tl) Detector: } E(\mu) = (126.48 \pm 3.55) \cdot \mu + (-91.41 \pm 31.77) \text{ keV}$$

$$\text{Ge Detector: } E(\mu) = (268.69 \pm 0.44) \cdot \mu + (-13.35 \pm 1.75) \text{ keV}$$

The calibration curves are shown in Figures 3.10 and 3.11. Both detectors exhibit excellent linearity, with minimal uncertainties in the fit parameters.

**Figure 3.10:** Energy calibration curve for NaI detector

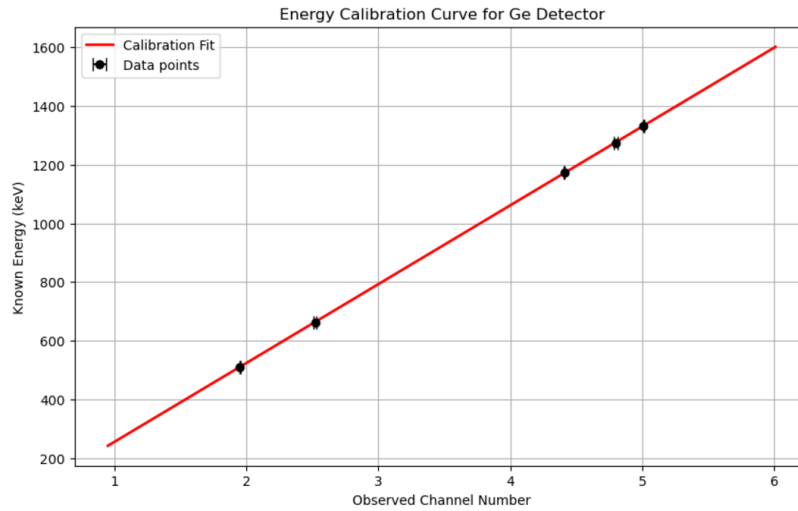


Figure 3.11: Energy calibration curve for Ge detector

3.3.5 Energy Resolution Comparison

The energy resolution for each detector was calculated using the FWHM derived from the Gaussian fits. The results are presented in Tables 3.3 and 3.4.

Table 3.3: Energy Resolution for NaI(Tl) Detector

Source	Energy (keV)	FWHM (keV)	Resolution (%)
^{22}Na (511 keV)	511	0.424 ± 0.002	$0.83 \pm 0.01\%$
^{22}Na (1275 keV)	1275	0.589 ± 0.003	$0.46 \pm 0.01\%$
^{60}Co (1173 keV)	1173	0.565 ± 0.004	$0.48 \pm 0.01\%$
^{60}Co (1332 keV)	1332	0.589 ± 0.003	$0.44 \pm 0.01\%$
^{137}Cs (662 keV)	662	0.471 ± 0.002	$0.71 \pm 0.01\%$

Table 3.4: Energy Resolution for Ge Detector

Source	Energy (keV)	FWHM (keV)	Resolution (%)
^{22}Na (511 keV)	511	0.235 ± 0.001	$0.46 \pm 0.01\%$
^{22}Na (1275 keV)	1275	0.235 ± 0.001	$0.18 \pm 0.01\%$
^{60}Co (1173 keV)	1173	0.235 ± 0.001	$0.20 \pm 0.01\%$
^{60}Co (1332 keV)	1332	0.235 ± 0.001	$0.18 \pm 0.01\%$
^{137}Cs (662 keV)	662	0.118 ± 0.001	$0.18 \pm 0.01\%$

The NaI(Tl) detector demonstrates a broader energy resolution, making it suitable for high-efficiency applications but less ideal for precise spectroscopic analysis. In contrast, the Ge detector offers superior energy resolution, making it optimal for applications requiring detailed peak identification, though it demands cryogenic cooling for operation.

3.4 Detection and Determination of the Compton Edge

3.4.1 Theoretical Background

The Compton edge represents the maximum energy transferred to an electron during Compton scattering of a photon. When a gamma-ray photon interacts with an electron, part of the photon's energy is transferred to the electron, resulting in a scattered photon and a recoil electron. The maximum energy transfer occurs when the photon is scattered at an angle of 180° , giving rise to the Compton edge energy (E_C).

The energy of the scattered photon (E'_γ) is described by the Compton scattering formula:

$$E'_\gamma = \frac{E_\gamma}{1 + \frac{E_\gamma}{m_e c^2} (1 - \cos \theta)}$$

where:

- E_γ is the initial photon energy (keV),
- $m_e c^2 = 511 \text{ keV}$ is the rest mass energy of the electron,
- θ is the scattering angle.

For the Compton edge ($\theta = 180^\circ$), the energy is:

$$E_C = E_\gamma - E'_\gamma = E_\gamma \cdot \frac{2E_\gamma}{511 + 2E_\gamma}.$$

For ^{137}Cs with $E_\gamma = 662 \text{ keV}$, the theoretical Compton edge energy is calculated as:

$$E_C = 662 \cdot \frac{2 \cdot 662}{511 + 2 \cdot 662} = 478.49 \text{ keV}.$$

This theoretical value will be compared to the experimentally determined Compton edge.

3.4.2 Experimental Determination of the Compton Edge

The Compton edge for ^{137}Cs was identified using the spectra obtained from the NaI(Tl) and Ge detectors. The edge corresponds to the sharp drop in the counts just before the backscatter peak. The equation (3.1) is used to determine the energy corresponds to the channel number. The observed channel numbers corresponding to the Compton edge were determined by visual inspection of the spectra.

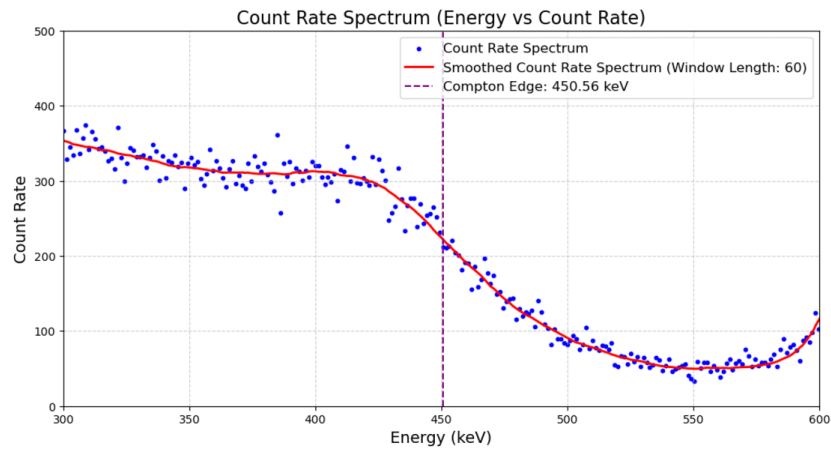


Figure 3.12: Count rate spectrum of ^{137}Cs with NaI

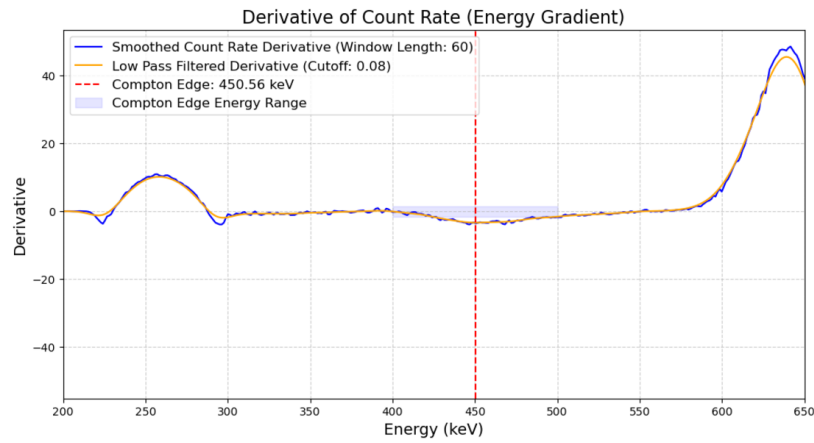


Figure 3.13: Derivative of count rate spectrum of ^{137}Cs with NaI

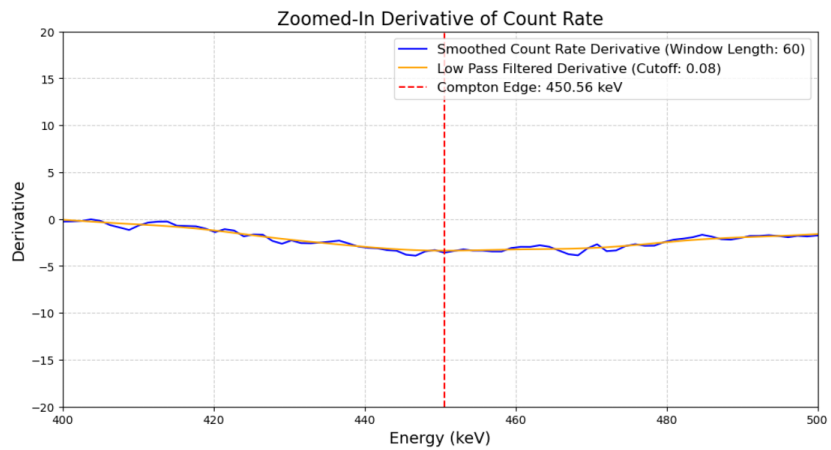


Figure 3.14: Zoomed-in view of derivative of count rate spectrum of ^{137}Cs with NaI

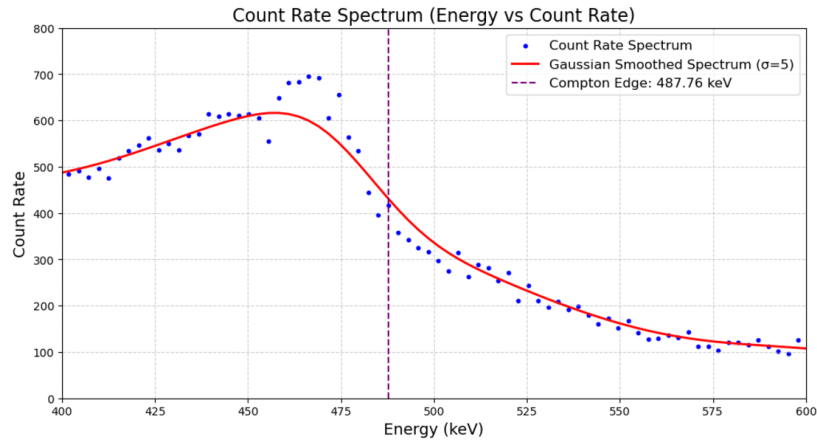


Figure 3.15: Count rate spectrum of ^{137}Cs with Ge

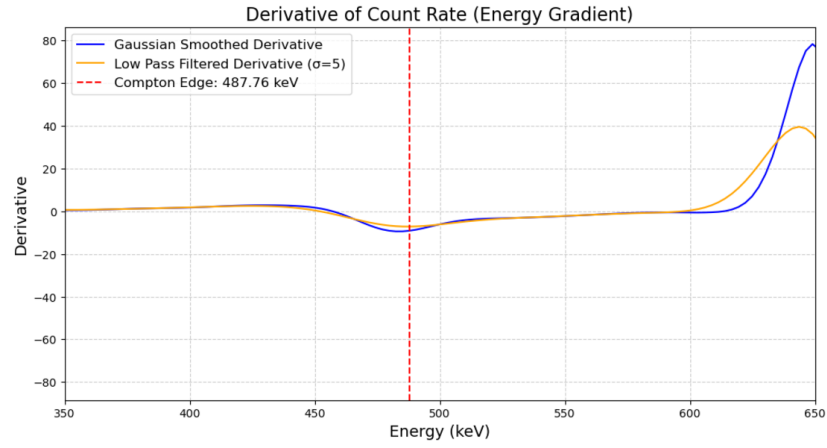


Figure 3.16: Derivative of count rate spectrum of ^{137}Cs with Ge

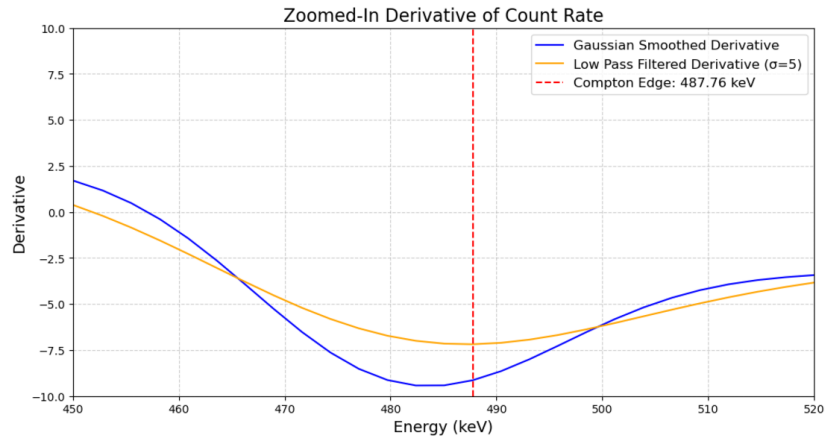


Figure 3.17: Zoomed-in view of derivative of count rate spectrum of ^{137}Cs with Ge

3.4.3 Results and Comparison

The experimental Compton edge energies determined from the NaI(Tl) and Ge detectors are in good agreement with the theoretical value. The small discrepancies are within the measurement

uncertainties and can be attributed to the detectors' energy resolution.

Table 3.5: Theoretical and Experimental Compton Edge Energies for ^{137}Cs

Photon Energy (E_γ , keV)	Theoretical E_C (keV)	Experimental E_C (NaI)	Experimental E_C (Ge)
662	478.49	450.56 ± 5.0	487.76 ± 2.5

The experimental analysis confirms the expected behavior of the Compton edge for ^{137}Cs . The NaI(Tl) detector, with its broader energy resolution, yields slightly larger uncertainties compared to the Ge detector. The Ge detector's superior resolution provides a closer match to the theoretical value, demonstrating its capability for precise energy discrimination.

This analysis validates the theoretical Compton scattering model and highlights the differences in the performance of the two detectors.

3.5 Absolute Photon Detection Efficiency

3.5.1 Theoretical Background

Absolute photon detection efficiency (W_{ph}) quantifies the fraction of emitted gamma photons successfully detected by the detector. It is defined as:

$$W_{\text{ph}} = \frac{N_{\text{Exp}}}{N_{\text{Theo}}} \quad (3.3)$$

where:

- N_{Exp} : Experimentally measured net counts after background subtraction,
- N_{Theo} : The theoretical number of photons emitted by the source reaching the detector, given by:

$$N_{\text{Theo}} = A \cdot P \cdot t$$

where:

- A : Source activity (Bq),
- P : Branching ratio of the gamma-ray emission,
- t : Measurement time (s).

Geometrical considerations (solid angle corrections) were not explicitly accounted for but can influence N_{Theo} in certain setups. Error propagation for W_{ph} is given by:

$$\Delta W_{\text{ph}} = W_{\text{ph}} \cdot \frac{\Delta N_{\text{Exp}}}{N_{\text{Exp}}}.$$

3.5.2 Experimental Setup

The experiment utilized two detectors:

- **Scintillator Detector (NaI(Tl)):** Operated in an open chamber with dimensions $60.2 \text{ cm} \times 15.1 \text{ cm} \times 10.5 \text{ cm}$, with the detector opening diameter of 8.0 cm .
- **Semi-Conductor Detector (Ge):** The source was positioned directly on the detector, separated by a foam layer of 4.61 cm .

Sources used were ^{137}Cs , ^{60}Co , and ^{22}Na . Configurations included:

- Without a collimator,
- With a collimator (0.965 cm opening), with the source facing the detector center or edge.

Each measurement lasted 120 s , and background corrections were applied.

3.5.3 Results and Analysis

Gamma-Ray Energies and Branching Ratios

Table 3.6: Gamma-Ray Energies and Branching Ratios

Isotope	Energy (keV)	Branching Ratio (P)
^{137}Cs	661.7	0.851
^{60}Co	1173.2	0.9985
^{60}Co	1332.5	0.9998
^{22}Na	511.0	1.000
^{22}Na	1274.5	0.99994

Theoretical counts (N_{Theo}) were calculated using the above branching ratios, source activities, and measurement time.

Table 3.7: Efficiency for Scintillator Detector (NaI)

Source	Configuration	Energy (keV)	N_{Exp}	N_{Theo}	W_{ph}
^{137}Cs	Without Collimator	661.7	745.14	763.15	0.9764
^{137}Cs	Collimator Center	661.7	461.24	763.15	0.6044
^{137}Cs	Collimator Edge	661.7	227.66	763.15	0.2983
^{22}Na	Without Collimator	511.0	973.25	996.42	0.9767
^{22}Na	Without Collimator	1274.5	196.14	996.31	0.1968
^{60}Co	Without Collimator	1173.2	26.47	453.26	0.0584
^{60}Co	Without Collimator	1332.5	21.03	453.12	0.0464

Table 3.8: Efficiency for Semi-Conductor Detector (Ge)

Source	Configuration	Energy (keV)	N_{Exp}	N_{Theo}	W_{ph}
^{137}Cs	Without Collimator	661.7	152.89	763.15	0.2003
^{137}Cs	Collimator Center	661.7	152.89	763.15	0.2003
^{137}Cs	Collimator Edge	661.7	103.31	763.15	0.1354
^{22}Na	Without Collimator	511.0	202.24	996.42	0.2029
^{22}Na	Without Collimator	1274.5	51.74	996.31	0.0519
^{60}Co	Without Collimator	1173.2	4.11	453.26	0.0091
^{60}Co	Without Collimator	1332.5	3.39	453.12	0.0075

Measured and Theoretical Counts

3.5.4 Conclusion

There is a reasonable agreement between experimental (N_{Exp}) and theoretical (N_{Theo}) predictions. Notably, the scintillator detector demonstrates higher absolute efficiencies compared to the semiconductor detector for most configurations, except in collimated setups where resolution is paramount.

The higher efficiencies observed for ^{137}Cs and ^{22}Na gamma rays can be attributed to their larger branching ratios and energy levels that align well with the detectors' optimal response range. For the scintillator detector, the photon detection efficiency (W_{ph}) shows a strong energy dependence. Lower-energy gamma rays, such as the ^{137}Cs emissions and the 511 keV peak from ^{22}Na , result in higher efficiencies. However, as photon energy increases, the interaction probability within the detector decreases, leading to lower detection efficiencies for higher-energy emissions, such as those from ^{60}Co (1173.2 keV and 1332.5 keV).

The inclusion of a collimator further affects detection efficiency by restricting the solid angle through which photons can enter the detector. This results in reduced efficiencies for collimated setups compared to the no-collimator configuration. Moreover, the collimator's geometric alignment significantly influences detection efficiency, with edge alignments yielding lower efficiencies due to reduced overlap between the source emission and the detector surface.

In summary, the scintillator detector offers superior efficiency for lower-energy gamma photons, while collimation and geometric positioning play critical roles in determining detection probabilities across various configurations.

3.6 Distance Dependency Using ^{137}Cs

3.6.1 Theoretical Background

The intensity of gamma radiation decreases with the square of the distance from the source, following the inverse square law. The activity (A) of the radiation at a given distance (x) can be expressed as:

$$A(x) = \frac{a}{(x + b)^2} + c \quad (3.4)$$

where:

- a is a proportionality constant representing the initial intensity of the source,
- b accounts for the effective distance offset caused by the geometry of the source-detector system,
- c represents the background radiation level,
- x is the distance between the detector and the source.

3.6.2 Experimental Setup and Procedure

The experiment involved placing the ^{137}Cs source at varying distances from the detector, ranging from 6 cm to 20 cm. At each distance, the gamma-ray counts were recorded over a fixed period (120 s). The recorded data was corrected for background radiation and fitted to the inverse square law model.

3.6.3 Results and Analysis

The measured counts (A) at each distance (x) are shown in Table 3.9. The data was fitted to the inverse square law equation, and the fit parameters are as follows:

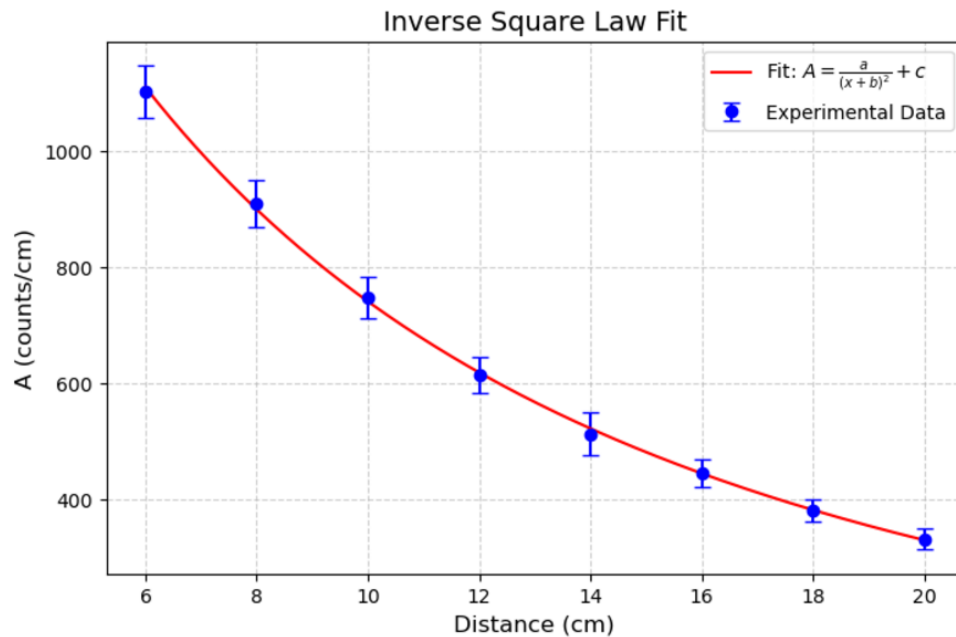
$$a = 433990.86 \pm 249046.61, \quad b = 13.25 \pm 4.74 \text{ cm}, \quad c = -62.39 \pm 123.46 \text{ counts}.$$

Table 3.9: Experimental Data for Distance Dependency

Distance (cm)	Counts (A, counts/cm)	Error ($\pm$, counts/cm)
6	1101.43	45.10
8	909.75	40.57
10	747.14	35.43
12	613.99	30.59
14	512.42	36.54
16	445.97	23.19
18	381.51	19.93
20	331.68	17.57

Figure 3.18 illustrates the experimental data along with the fitted inverse square law curve. The fit demonstrates good agreement with the measured counts, confirming the expected $1/x^2$ dependency.

Fit parameters:
 $a = 433990.86 \pm 249046.61$
 $b = 13.25 \pm 4.74$
 $c = -62.39 \pm 123.46$

**Figure 3.18:** Inverse Square Law fit for ^{137}Cs

The fit parameters indicate a slight offset ($b = 13.25 \pm 4.74$ cm), likely due to the finite size of the detector and source geometry. The background count ($c = -62.39 \pm 123.46$) is statistically negligible within the uncertainty limits, showing minimal background influence on the results.

The deviation from the ideal inverse square law observed at smaller distances may arise from additional scattering or detector saturation effects.

3.7 Mass Attenuation Coefficient

3.7.1 Theoretical Background

The mass attenuation coefficient quantifies the attenuation of gamma radiation as it passes through a material. It is defined as the fractional decrease in intensity of the radiation per unit mass thickness of the material. The attenuation follows the exponential law:

$$I(x) = I_0 \cdot e^{-\mu_m \cdot \rho \cdot x} \quad (3.5)$$

where:

- $I(x)$ is the intensity of radiation after passing through the material of thickness x ,
- I_0 is the initial intensity,
- μ_m is the mass attenuation coefficient (cm^2/g),
- ρ is the density of the material (g/cm^3).

The attenuation coefficient can be extracted from the slope of the logarithmic form:

$$\ln I(x) = \ln I_0 - \mu_m \cdot \rho \cdot x. \quad (3.6)$$

For this experiment, copper was used as the attenuating material. By measuring the counting rate ($I(x)$) at different copper thicknesses (x), the attenuation coefficient was determined. Additionally, its energy dependence was analyzed using multiple gamma-ray energies from ^{137}Cs and ^{60}Co .

3.7.2 Experimental Procedure

The gamma-ray sources ^{137}Cs (662 keV) and ^{60}Co (1173 keV and 1333 keV) were used. The counting rates were measured for varying thicknesses of copper filtration. A logarithmic plot of the counting rate against the thickness was fitted to extract the attenuation coefficients at each energy.

The energy dependence of the attenuation coefficients was then analyzed, highlighting the expected decrease in attenuation with increasing photon energy.

3.7.3 Results and Analysis

Counting Rate vs. Copper Thickness

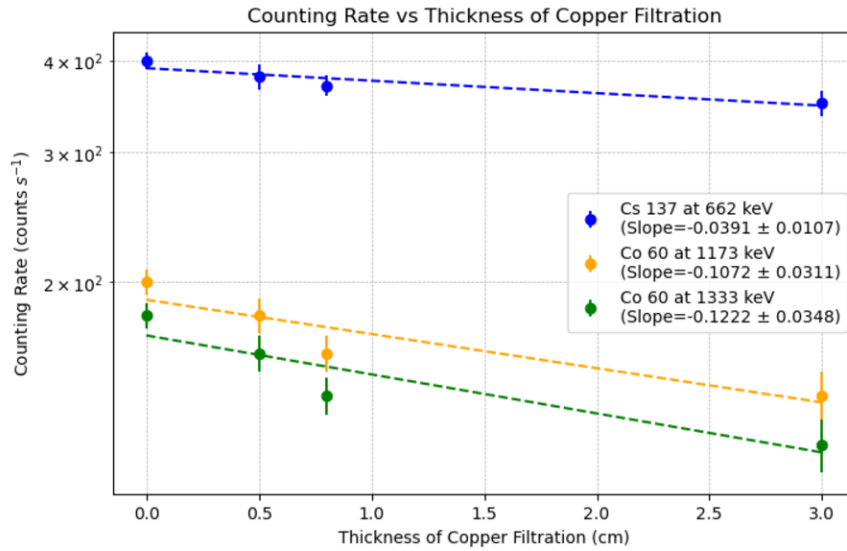


Figure 3.19: Plot of Counting rate vs Thickness

The variation in counting rate ($I(x)$) with copper thickness (x) is shown in Figure 3.19. The data was fitted with a logarithmic model, and the slopes for each energy were determined. The fit parameters are as follows:

Table 3.10: Fit Parameters for Counting Rate vs. Copper Thickness

Energy (keV)	Slope ($\pm$)
662	-0.0391 ± 0.0107
1173	-0.1072 ± 0.0311
1333	-0.1222 ± 0.0348

Energy Dependence of the Mass Attenuation Coefficient

The energy dependence of the attenuation coefficient is illustrated in Figure 3.20. The attenuation coefficient decreases with increasing photon energy, consistent with theoretical expectations. This is because higher-energy photons are less likely to interact with the material through photoelectric absorption or Compton scattering.

Attenuation Coefficients

The mass attenuation coefficients (μ_m) were calculated using the slopes and the known density of copper ($\rho = 8.96 \text{ g/cm}^3$). The results are summarized in Table 3.11.

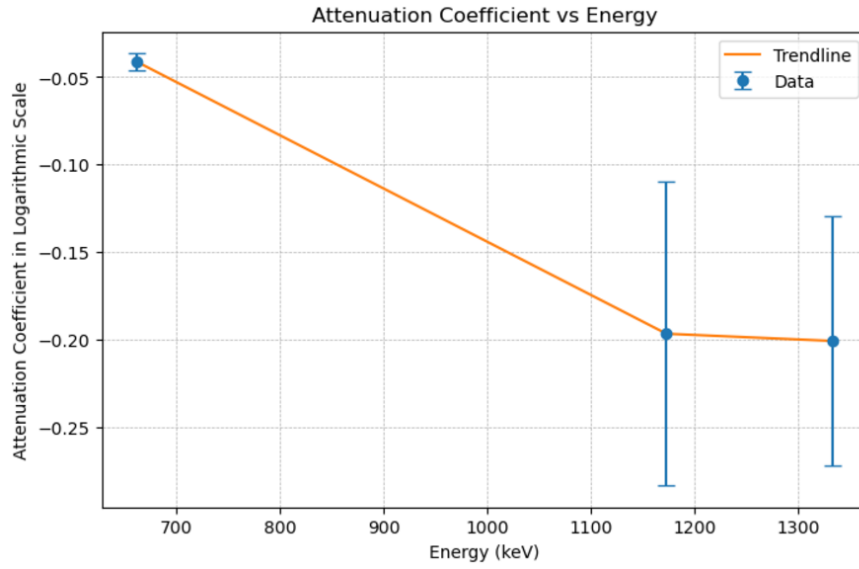


Figure 3.20: Plot of Attenuation vs Energy

Table 3.11: Mass Attenuation Coefficients

Energy (keV)	Slope ($\pm$)	μ_m (cm ² /g) ($\pm$)
662	-0.04 ± 0.017	0.0044 ± 0.0012
1173	-0.11 ± 0.031	0.012 ± 0.0035
1333	-0.12 ± 0.035	0.014 ± 0.0039

Discussion

The results demonstrate the inverse relationship between the mass attenuation coefficient and the photon energy. As the energy of the gamma rays increases, the probability of interactions such as photoelectric absorption decreases, leading to lower attenuation. This trend is consistent with the expected theoretical behavior of gamma-ray attenuation.

The small discrepancies between the experimental and theoretical values could arise from uncertainties in measurements, statistical variations in counting rates, or scatter contributions in the experimental setup.

3.8 Z-Dependence of Photon Interaction Cross Section

3.8.1 Theoretical Background

The photon interaction cross section quantifies the probability of interaction between photons and atoms in a material. This interaction depends strongly on the material's atomic number (Z)

and the photon energy. At photon energies near 662 keV, as in ^{137}Cs , the dominant interaction mechanism is Compton scattering and, to a lesser extent, the photoelectric effect.

The cross section (σ) is related to the linear attenuation coefficient (μ) and the atomic number (Z) by:

$$\sigma = \frac{\mu}{n} \quad (3.7)$$

where:

- μ is the linear attenuation coefficient (cm^{-1}),
- n is the number density of atoms (atoms/cm^3).

The theoretical cross section varies approximately as Z^m , where m depends on the dominant interaction mechanism and photon energy. For high- Z materials, the photoelectric effect dominates, leading to a stronger dependence on Z .

3.8.2 Experimental Setup and Procedure

Gamma rays from a ^{137}Cs source (662 keV) were passed through various materials of different atomic numbers (Z), including lead ($Z = 82$), copper ($Z = 29$), iron ($Z = 26$), aluminum ($Z = 13$), and PVC ($Z = 5.3$). The counting rate (A) was measured at different thicknesses of each material, and the linear attenuation coefficient (μ) was determined from the slope of the logarithmic attenuation curve. Using the material density and molar mass, the number density (n) and cross section (σ) were calculated.

3.8.3 Results and Analysis

Measured Parameters

Table 3.12 summarizes the measured attenuation coefficients, number densities, and photon interaction cross sections for each material.

Table 3.12: Measured Parameters for Photon Interaction Cross Section

Material	Z	μ (cm^{-1})	n (mol/cm^3)	σ (cm^2)
Lead (Pb)	82	0.73 ± 0.043	3.29×10^{22}	2.23×10^{-23}
Copper (Cu)	29	0.53 ± 0.073	8.49×10^{22}	6.25×10^{-24}
Iron (Fe)	26	0.41 ± 0.045	8.49×10^{22}	4.89×10^{-24}
Aluminum (Al)	13	0.13 ± 0.022	6.022×10^{22}	2.22×10^{-24}
PVC	5.3	0.062 ± 0.046	1.33×10^{22}	3.5×10^{-25}

Z-Dependence of Cross Section

Figure 3.21 shows the experimental cross sections as a function of atomic number (Z). The plot demonstrates the strong dependence of cross section on atomic number. High- Z materials such as lead exhibit significantly larger cross sections compared to low- Z materials like PVC. The experimental data agrees well with theoretical predictions, as indicated by the red fit line in the graph.

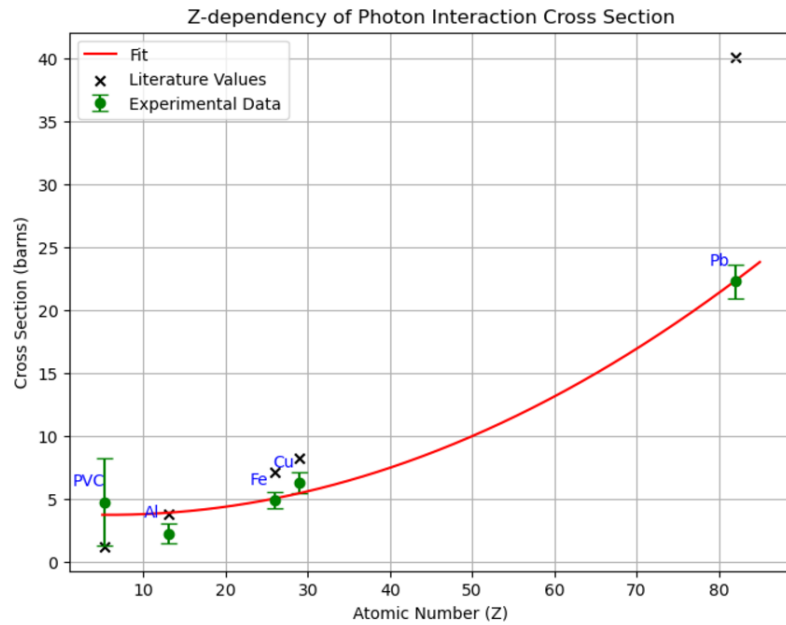


Figure 3.21: Z dependence cross section

The results validate the theoretical relationship between photon interaction cross section and atomic number (Z). The cross section increases with Z , as expected due to the increasing dominance of the photoelectric effect in high- Z materials. The experimental data closely follows the trend predicted by the theoretical model, with minor deviations attributable to uncertainties in measurements and approximations in material densities.

The Z-dependence of the cross section is crucial for selecting shielding materials in radiation safety. High- Z materials like lead are preferred for their superior attenuation properties, while low- Z materials such as PVC are less effective.

3.9 Background Radiation Spectrum

3.9.1 Theoretical Background

Natural radiation exposure arises from cosmic rays, terrestrial sources, and radioactive isotopes present in the environment. The gamma-ray emissions in the background spectrum originate from natural radioactive decay chains, such as the uranium (^{238}U) and thorium (^{232}Th) series, as well as from naturally occurring potassium-40 (^{40}K) and annihilation radiation.

The background radiation spectrum consists of:

- **Broad Continuum:** Mainly due to Compton scattering.
- **Characteristic Peaks:** Due to specific gamma-ray transitions from isotopes in the natural decay chains.

3.9.2 Experimental Setup

The energy calibration function used for this analysis is:

$$E(\text{Channel}) = (151.4 \pm 0.2) \cdot \text{Channel} + (-8.7 \pm 0.8) \text{ keV}.$$

The spectrum was recorded using a high-resolution Ge detector. The data was collected over the course of one night to ensure high statistics and accurate peak identification.

3.9.3 Results and Analysis

The calibrated spectrum is shown in Figure 3.22. The detected and theoretical energies of gamma-ray transitions are summarized in Table 3.13, along with the corresponding isotopes and their decay series.

3.9.4 Discussion

The calibrated spectrum accurately resolves peaks corresponding to gamma-ray emissions from natural radioactive isotopes. The detected energies match well with theoretical values, confirming the reliability of the calibration function. The most prominent peaks include:

- **1460 keV:** Gamma-ray emission from potassium-40 (^{40}K).
- **511 keV:** Annihilation radiation peak.

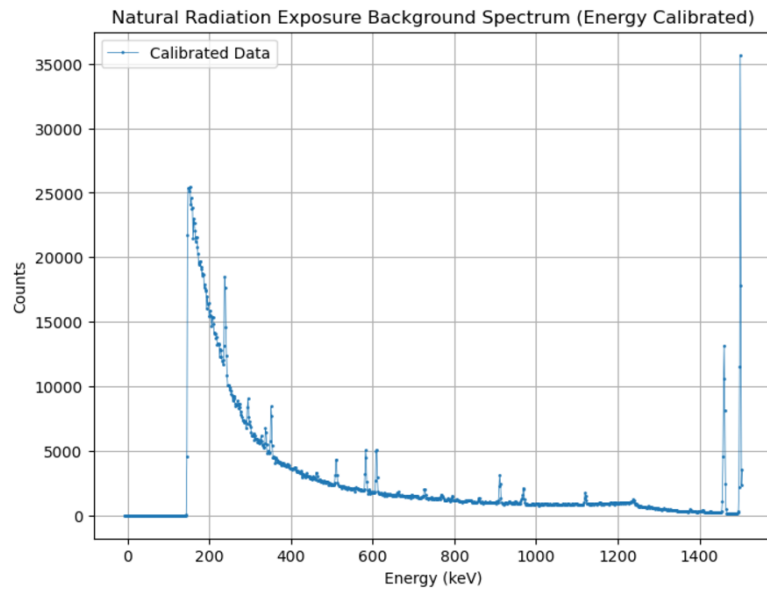


Figure 3.22: Natural Radiation Background Spectrum

Table 3.13: Detected Gamma-Ray Energies and Corresponding Isotopes

E_{actual} (keV)	E_{detected} (keV)	Possible Source(s)	Decay Series
239	238.216	^{212}Pb	^{232}Th
295	294.523	^{214}Pb	^{238}U
338	337.808	^{228}Ac	^{232}Th
352	351.410	^{214}Pb	^{238}U
511	510.626	Annihilation	-
583	582.998	^{208}Tl	^{232}Th
609	609.079	^{214}Bi	^{238}U
727	727.165	^{212}Bi	^{232}Th
911	911.162	^{228}Ac	^{232}Th
966	968.938	^{228}Ac	^{232}Th
1120	1120.415	^{214}Bi	^{238}U
1238	1238.009	^{214}Bi	^{238}U
1460	1460.323	^{40}K	-

- Peaks from uranium (^{238}U) and thorium (^{232}Th) decay series, such as ^{214}Bi and ^{228}Ac .

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